Ridwan Ademola TAIWO

Kowloon, Hong Kong

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CAREER OBJECTIVES

To leverage my expertise in artificial intelligence and construction engineering to advance intelligent built environments through innovative research in explainable AI systems, smart wearable technologies for construction safety, and sustainable infrastructure management. I seek a position where I can lead interdisciplinary research teams, translate academic innovations into industry-ready solutions, and establish a research program that bridges AI technology with practical construction applications. My long-term vision is to contribute significantly to creating safer, more sustainable, and efficient infrastructure systems while mentoring the next generation of researchers and maintaining active industry engagement to address global challenges in urbanization and climate change.

EDUCATION

The Hong Kong Polytechnic University, Hong Kong (Ranked 57th globally, QS 2025)
Ph.D. in Construction and Infrastructure Engineering (01/2022 – 2/2025)

- Achieved a CGPA of 4.0/4.3 (Distinction)
- Published 7 high-impact peer-reviewed articles and 1 currently under-review from my Ph.D. research
- Served as an academic guest at ETH Zurich for a six-month period

The University of Cape Town, South Africa
MEng. in Civil Infrastructure Management & Maintenance with Distinction (01/2020 – 12/2021)

- Specialized in Construction Management and Construction Materials
- Graduated with 79.13% (Top 1% of my cohort)

The Federal University of Technology, Akure, Nigeria
BEng. in Civil Engineering with First Class Honours (09/2013 – 11/2018)

- Graduated with a CGPA of 4.60/5.00 (Top 1% of my cohort)
- Achieved a placement on the Dean's list for 9 out of 10 semesters

RESEARCH INTERESTS

Multidisciplinary research interests include:

- Construction safety
- Smart wearable technologies
- Digital and sustainable construction
- Explainable AI systems
- AI in built environments

RESEARCH EXPERIENCE

Department of Building and Real Estate, the Hong Kong Polytechnic University
Research Associate (03/2024 – present)

- Conducting research on the integration of generative AI, including large language models, within the construction industry.
- Developing and refining research proposals, securing grants, and managing project budgets and timelines.
- Providing academic guidance and mentorship to bachelor's and master's students, supporting their research and professional development.
- Offering technical expertise and research assistance to PhD candidates.

Department of Building and Real Estate, the Hong Kong Polytechnic University
Research Assistant (01/2022 – 2/2025)

- Led the development of generative AI models to enhance productivity in the civil engineering industry, employing project management techniques to guide a team of junior researchers.
- Investigated strategies for circular economy adoption in the construction industry.
- Developed innovative research ideas and prepared grant applications, including project timelines and budget management.
- Assisted in organizing workshops and seminars to communicate research findings to industry experts.
- Supervised and provided technical support for bachelor’s and master’s students, mentoring students to successful completion.
- Provided technical support for PhD students.

Institute of Construction and Infrastructure Management, Department of Civil, Environmental, and Geomatic Engineering, ETH Zurich
Academic Guest (04/2024 - 10/2024)

- Conducted innovative research on explainable predictive maintenance for highways to enhance performance, safety, and sustainability.
- Conducted research to investigate the causes of water pipe failure, utilizing advanced analytical techniques to develop more reliable and efficient solutions for infrastructure management.
- Developed an end-to-end framework for predicting the strength of high-performance concrete, incorporating construction management methodologies.

Afrikan Research Initiative, South Africa (Part-time volunteering service)
Research Associate (02/2020 – 12/2022)

- Contributed to a team involved in modeling a risk assessment model for Covid-19 in African countries.
- Participated in data collection and analysis to inform evidence-based decision-making.

JOURNAL ARTICLES

1. **Taiwo, R.*,** Zayed, T., Bakhtawar, B., & Adey, B. T. (2025). “Explainable deep learning models for predicting water pipe failures” Journal of Environmental Management (IF = 8.0, **Q1*****, top 90th percentile) 379, 124738. https://doi.org/10.1016/j.jenvman.2025.124738
2. **Taiwo, R.*,** Shaban, I. A., Ahmad, T. & Zayed, T. (2025). “Big data-driven prediction of watermain failures in semi-tropical regions: Case study of Hong Kong's distribution network” Automation in Construction (IF = 9.6, **Q1***,** top 96th percentile) 175, 106159. https://doi.org/10.1016/j.autcon.2025.106159
3. **Taiwo, R.*,** Yussif, A.-M., & Zayed, T. (2025). "Making waves: Generative artificial intelligence in water distribution networks: Opportunities and challenges" Water Research X (IF = 7.2, **Q1***,** top 97th percentile), 28, 100316. https://doi.org/10.1016/j.wroa.2025.100316
4. **Taiwo, R.*,** Bello, I. T., Abdulai, S. F., Yussif, A.-M., Salami, B. A., Saka, A., Ben Seghier, M. E. A., & Zayed, T. (2025). "Generative Artificial Intelligence in Construction: A Delphi Approach,

Framework, and Case Study." Alexandria Engineering Journal (IF = 6.2, **Q1***,** top 95th percentile), 116, 672-698. https://doi.org/10.1016/j.aej.2024.12.079

5. **Taiwo, R.*,** Zayed, T., Elshaboury, N., Alfalah, G., & Abdelkader, E. M. (2025). "Promoting sustainable water distribution networks: Modeling of water pipe failure factors and modes." Cleaner Engineering and Technology (IF = 5.3, **Q1*****, top 80th percentile), 26, 100969. https://doi.org/10.1016/j.clet.2025.100969
6. Yussif, A., **Taiwo, R.,** Zayed, T., Mohandes, S.R., & Abdulai, S. F. (2025). "Enhancing Sustainable Urban Mobility: Computer Vision Applications in Pedestrian Path Studies". Journal of Traffic and Transportation Engineering (IF = 7.4, **Q1***,** top 97th percentile), in Press. https://jtte.chd.edu.cn/article/id/759eddae-7ad8-43b1-8039-85ec46f1119a
7. Adedara, M. L., **Taiwo, R.,** Ayeleru, O. O., & Bork, H.-R. (2025). "Evaluating recycling initiatives for landfill diversion in developing economies using integrated machine learning techniques." Recycling (IF = 4.6, Q2, top 57th percentile), 10(3), 100. https://doi.org/10.3390/recycling10030100
8. **Taiwo, R.*,** Yussif, A., Adegoke, A.H., & Zayed, T. (2024). "Prediction and deployment of compressive strength of high-performance concrete using ensemble learning techniques". Construction and Building Materials (IF = 7.4, **Q1***,** top 97th percentile), 451, 138808. https://doi.org/10.1016/j.conbuildmat.2024.138808
9. **Taiwo, R.*,** Yussif, A., Ben Seghier, M. E. A., & Zayed, T. (2024). "Explainable Ensemble Models for Predicting Wall Thickness Loss of Water Pipes". Ain Shams Engineering Journal (IF = 6.0, **Q1***,** top 95th percentile), https://doi.org/10.1016/j.asej.2024.102630
10. **Taiwo, R.*,** Zayed, T. & Ben Seghier, M. E. A. (2024). "Integrated intelligent models for predicting water pipe failure probability". Alexandria Engineering Journal (IF = 6.2, **Q1***,** top 95th percentile), 86, 243-257 https://doi.org/10.1016/j.aej.2023.11.047
11. Muddassir, M., Zayed, T, **Taiwo, R.,** & Ben Seghier, M. E. A. (2024). "Advancing the analysis of water pipe failures: a probabilistic framework for identifying significant factors". Scientific Reports (IF = 3.8, **Q1***,** top 82nd percentile), 14, 19218 https://doi.org/10.1038/s41598-024-69855-w
12. Akindele, N., **Taiwo, R.,** Sarvari, H., Oluleye, B. I., Awodele, I. A., & Olaniran, T. O. (2024). "A state-of-the-art analysis of virtual reality applications in construction health and safety". Results in Engineering (IF = 6.0, **Q1*****, top 95th percentile), 23, 102382 https://doi.org/10.1016/j.rineng.2024.102382
13. Abdulai, S. F., Nani, G., **Taiwo, R.,** Antwi-Afari, P., Zayed, T., & Sojobi, A. O. (2024). "Modelling the relationship between circular economy barriers and drivers for sustainable construction industry." Building and Environment (IF = 7.1 **Q1*****, top 93th percentile), 254, 111388 https://doi.org/10.1016/j.buildenv.2024.111388
14. Yussif, A.-M., Zayed, T., **Taiwo, R.,** & Fares, A. (2024). "Promoting sustainable urban mobility via automated sidewalk defect detection." Sustainable Development (IF = 9.9, **Q1***,** top 99th percentile), [1-21] https://doi.org/10.1002/sd.2999
15. Saka, A., **Taiwo, R.,** Saka, N., Salami, B. A., Ajayi, S., Akande, K., & Kazemi, H. (2024). "GPT models in construction industry: Opportunities, limitations, and a use case validation." Developments in the Built Environment (IF = 6.2, **Q1***,** top 93th percentile), 17, 100300. https://doi.org/10.1016/j.dibe.2023.100300
16. Awodele, I.A., Chan, A.P.C., Mewomo, M.C., Darko, A., Municio, A.M.G., **Taiwo, R.,** Eze, E.C., Awodele, O.A., & Olatunde, N.A. (2024). "Awareness, adoption readiness and challenges of railway 4.0 technologies in a developing economy." Heliyon (IF = 3.4, **Q1***,** top 80th percentile), 10, e25934 https://doi.org/10.1016/j.heliyon.2024.e25934
17. Salami, B. A., Bahraq, A. A., Moin ul Haq, M., Ojelade, O. A., **Taiwo, R.,** Wahab, S., Adewumi, A. A., & Ibrahim, M. (2024). "Polymer-enhanced concrete: A comprehensive review of innovations and

pathways for resilient and sustainable materials." Next Materials, 4(100225). https://doi.org/10.1016/j.nxmate.2024.100225

18. Abdelkader, E. M., Zayed, T., Elshaboury, N., & **Taiwo, R.** (2024). "A hybrid Bayesian optimization-based deep learning model for modeling the condition of saltwater pipes in Hong Kong." International Journal of Construction Management (IF = 3.4, **Q1*****, top 75th percentile), https://doi.org/10.1080/15623599.2024.2304392
19. Bello, I. T., **Taiwo, R.,** Esan, O. C., Adegoke, A. H., Ijaola, A. O., Li, Z., Zhao, S., Wang, C., Shao, Z., & Ni, M. (2024). "AI-enabled materials discovery for advanced ceramic electrochemical cells." Energy and AI (IF = 9.6, **Q1*****, top 93rd percentile), 15, 100317. https://doi.org/10.1016/j.egyai.2023.100317
20. Farh, H.M.H, Ben Seghier, M. E. A, **Taiwo, R.,** & Zayed, T. (2023). "Analysis and Ranking of Corrosion Causes for Water Pipelines: A Critical Review." npj Clean Water (IF = 10.4, **Q1***,** top 99th percentile), 6, 65 https://doi.org/10.1038/s41545-023-00275-5
21. **<u>Taiwo, R.</u>*,** Shaban, I. A., & Zayed, T. (2023). "Development of sustainable water infrastructure: A proper understanding of water pipe failure". Journal of Cleaner Production (IF = 9.7, **Q1***,** top 93rd percentile), 398: 136653 https://doi.org/10.1016/j.jclepro.2023.136653
22. **<u>Taiwo, R.</u>*,** Ben Seghier, M. E. A., & Zayed, T. (2023). "Towards sustainable water infrastructure: The state-of-the-art for modeling the failure probability of water pipes". Water Resources Research (IF = 4.6, **Q1***,** top 93rd percentile). e2022WR033256. https://doi.org/10.1029/2022WR033256
23. Adedara, M. L., **Taiwo, R.,** & Bork, H.-R. (2023). "Municipal Solid Waste Collection and Coverage Rates in Sub-Saharan African Countries: A Comprehensive Systematic Review and Meta-Analysis." Waste, 1(2), 389-413. https://doi.org/10.3390/waste1020024
24. **<u>Taiwo, R.</u>*,** Hussein, M., & Zayed, T. (2022). "An Integrated Approach of Simulation and Regression Analysis for Assessing Productivity in Modular Integrated Construction Projects." Buildings (IF = 3.1, Q2, top 72nd percentile), 12: 2018. https://doi.org/10.3390/buildings12112018
25. Dolphin, W.S., Alshami, A.A., Tariq, S., Boadu. V., Mohandes, S.R., **Taiwo, R.,** & Zayed, T. (2021). "Effectiveness of policies and difficulties in improving safety performance of repair, maintenance, minor alteration, and addition works in Hong Kong". International Journal of Construction Management. (IF = 3.4, **Q1***,** top 75th percentile) https://doi.org/10.1080/15623599.2021.1935130
26. Yeung, H.C., **Taiwo, R.,** Tariq, S. & Zayed, T. (2020). "BEAM Plus implementation in Hong Kong: assessment of challenges and policies." International Journal of Construction Management (IF = 3.4, **Q1***,** top 75th percentile). https://doi.org/10.1080/15623599.2020.1827692

*** First authored articles = 11**

- Corresponding author articles = 11

***** JCR Q1 journal articles = 22**

REFEREED CONFERNCE PAPERS

1 **Taiwo, R.,** Ben Seghier, M. E. A., & Zayed, T. (2023). "Predicting Wall Thickness Loss in Water Pipes Using Machine Learning Techniques". In EuroStruct Conference (pp. 1-10). Boku, Austria. September 25-29. https://doi.org/10.1002/cepa.2075

2 **Taiwo, R.,** Wang, K.C.; Olanrewaju, O.I.; Tariq, S.; Abimbola, O.T.; Mehmood, I.; Zayed, T. (2022)." An Analysis of Employee Motivation in the Construction Industry: The Case of Hong Kong" Engineering Proceedings (Cite Score: 0.7) 22: 11 https://doi.org/10.3390/engproc2022022011

INDUSTRY REPORT

1 Zayed, T. et al. (2024). "Modeling and Analysis of Water Pipe Failure: Investigate Causes, Find Solutions, and Develop Potential Strategies, Polices, and Regulations". Department of Building and Real Estate, the Hong Kong Polytechnic University.

PREPRINT PAPERS

1 **Taiwo, R.,** Bello, I. T., Abdulai, S. F., Yussif, A.-M., Salami, B. A., Saka, A., & Zayed, T. (2024). "Generative AI in the Construction Industry: A State-of-the-art Analysis." arXiv (Preprint). https://doi.org/10.48550/arXiv.2402.099392
2 Saka, A., **Taiwo, R.,** Saka, N., Oluleye, B., Dauda, J., & Akanbi, L. (2024). "Integrated BIM and Machine Learning System for Circularity Prediction of Construction Demolition Waste." arXiv (Preprint). https://doi.org/10.48550/arXiv.2407.14847

INVITED PRESENTATIONS

1 23rd Scientific Seminar of the Group Mechanics and Construction, Institute of Computational Science & Artificial Intelligence, Van Lang University, Vietnam. Leveraging Machine Learning Techniques for Pipe Thickness Assessment. February 2024.
2 Annual Research Forum for the Association of Nigerian Scholars in Hong Kong. Assessing Productivity in Modular Integrated Construction Projects: An Integrated Approach of Simulation and Regression. June 2023.

RESEARCH GRANTS/PROJECTS

- Integrated Intelligent Models for Effective Management of Water Distribution Network. Grant Number: **G32R006**. Funding: **HKD 55,000**. Funder: **The Hong Kong Polytechnic University (PolyU Scholars)**. Duration: April – October 2024. Role: **Principal Investigator**
- Modeling and Analysis of Water Pipe Failure: Investigate Causes, Find Solutions, and Develop Potential Strategies, Policies, and Regulations. Grant Number: **ITS/033/20FP**. Funding: **HKD 7.6 million**. Funder: **Innovation and Technology Fund - Innovation and Technology Support Programme (ITF-ITSP)**. Duration: August 2021 – March 2024. Principal Investigator: Professor Tarek ZAYED. Role: **Researcher**
- Eco-friendly reusable pads. Grant Number: **SEF009**. Funding: **USD 3,000**. Funder: **Scholars Entrepreneurship Fund, MasterCard Foundation, Canada**. Duration: August 2021 – December 2023. Role: **Principal investigator**

TEACHING QUALIFICATION

Postgraduate Certificate in Teaching: Effective Teaching Assistant (BETA) (June 2022)
Education Development Centre, the Hong Kong Polytechnic University, Hong Kong

TEACHING PHILOSOPHY

My teaching philosophy centers on fostering a dynamic and inclusive learning environment that encourages critical thinking, active engagement, and a lifelong passion for knowledge. Positive feedback on my teaching engagements affirms this approach, highlighting my ability to empower students to become independent thinkers and problem-solvers. I emphasize interactive methods that cater to diverse learning styles, integrating technology and real-world applications. My goal is to cultivate an academic setting where students feel valued and motivated to explore and challenge ideas. Committed to continuous improvement, I reflect on and adapt my practices to meet the evolving needs of all learners.

TEACHING EXPERIENCE

Department of Building and Real Estate, the Hong Kong Polytechnic University, Hong Kong
Teaching Assistant (01/2022-02/2025)

- Tutored undergraduate and postgraduate modules on construction technology, civil engineering materials, and simulation and IT applications in construction, thereby providing in-depth understanding and guidance to students. **Average class size in these modules is 88.**
- Designed and developed innovative module materials, incorporating the latest industry practices and technological advancements.
- Marking of undergraduate scripts and invigilation of exams, ensuring fair evaluation and adherence to academic standards.
- Consistently receiving high student satisfaction ratings, averaging 4.5/5.0 in end-of-semester evaluations.
- Provided pastoral teaching support to students.

Department of Civil Engineering, University of Cape Town, South Africa
Teaching Assistant (02/2020-12/2021)

- Delivered tutorial lectures on structural and infrastructure engineering to undergraduate students, resulting in improved success rates in examinations. **Average class size in these modules is 115.**
- Contributed to the design and development of innovative module materials, integrating current industry practices and technological advancements.
- Supervised undergraduate exams and evaluated their scripts, ensuring accuracy and consistency in grading.
- Offered pastoral teaching support to undergraduate students

Department of Science, Manbahu College of Science and Islamic Studies
Instructor – part-time (09/2011-09/2018)

- Instructed secondary school students in Mathematics and Physics, fostering a comprehensive understanding of the subjects and promoting academic success.
- Developed lesson plans, teaching materials, and assessments to ensure effective knowledge transfer and student engagement.
- The success rate of the students in Senior Secondary Certificate Examination (SSCE) in Mathematics and Physics increased by **21 and 28%,** respectively on average.

TEACHING MODULES

The Hong Kong Polytechnic University, Hong Kong

- CE631(Level 8): Simulation and IT Applications in Construction
- BRE265 (Level 6): Construction Technology and Materials
- BRE272 (Level 6): Project Supervision and Contract Administration

University of Cape Town

- CIV4045F (Level 6): Structural Design

INDUSTRY EXPERIENCE

Webster Global Ventures Limited, Warri, Delta State, Nigeria
Site Engineer (03/2019-01/2020)

- Supervised the construction of rigid pavement and drains, effectively mitigating erosion risks in the Adidi community, Delta State.
- Assisted in the management of site materials and equipment, ensuring efficient operations.

MIT Nosmas Smith Limited, Ibadan, Oyo State, Nigeria
Assistant Site Engineer (05/2017 – 11/2017)

- Assisted in the coordination of various construction aspect of Hotel Diganga, Ile-Ife, Osun State, Nigeria.
- Prioritized site safety by diligently monitoring and enforcing health and safety regulations.

Slava Yeditepe Projects Limited, Osun State, Nigeria
Construction Supervisor (09/2016 – 12/2016)

- Part of the supervision team for a 33m long prestressed bridge that effectively resolved traffic congestion in Osun State
- Managed construction materials and manpower on-site, ensuring smooth project execution.

AWARDS AND HONORS

Research Student Attachment Program Scholarship, the Hong Kong Polytechnic University (2023)

- Awarded the scholarship (HKD 55,000) to pursue a six-month attachment program at ETH Zurich, Switzerland

ALX Africa Scholarship (2022)

- Awarded the scholarship (valued at $7500) to pursue a nanodegree course in data analytics

University Grant Council of Hong Kong, Research Postgraduate Scholarship (2021)

- Awarded the scholarship to pursue a Ph.D. degree at the Hong Kong Polytechnic University

Top 10 Winners for Baobab Virtual Summit (2020)

- Recognized as one of the top 10 winners for Baobab Virtual Summit organized by MasterCard Foundation.
- Demonstrated exceptional skills and knowledge in the summit's focus area.

Nawa Ignacy Lukasiewicza Scholarship for a Master's Degree in Poland (2020)

- Awarded the prestigious Nawa Ignacy Lukasiewicza Scholarship for pursuing a master's degree in Poland.

Global Excellence and Stature (GES) 4.0 Postgraduate Funding at University of Johannesburg (2020)

- Received the Global Excellence and Stature (GES) 4.0 Postgraduate Funding at the University of Johannesburg based on academic performance and research potential.

MasterCard Foundation Scholars Program for Postgraduate Studies at the University of Cape Town (2019)

- Selected as a recipient of the MasterCard Foundation Scholars Program for pursuing postgraduate studies at the University of Cape Town.
- Acknowledged for outstanding achievements and leadership qualities.

Second Best Student in 2018, Department of Civil Engineering, Federal University of Technology Akure, Nigeria (2018)

- Achieved the honor of being the second-best student in the Department of Civil Engineering at the Federal University of Technology Akure, Nigeria in 2018.
- Demonstrated exceptional academic performance and dedication to studies.

AbdulKabir Aliu Scholarship Foundation (2017-2018)

- Received undergraduate funding of $500USD from the AbdulKabir Aliu Scholarship Foundation.

Dean's List Placement in the School of Engineering, Federal University of Technology Akure, Nigeria (2013-2018)

- Consistently placed on the Dean's List in the School of Engineering at the Federal University of Technology Akure, Nigeria from 2013 to 2018.

SKILLS

Technical Skills

- Proficient in using machine learning and computer vision algorithms for predictive modeling and analysis.

- Proficient in using various generative AI algorithms and techniques, including large language models, to solve engineering problems.
- Skilled in data collection, cleaning, and analysis using statistical software such as SPSS, Stata, and Nvivo.
- Experienced in utilizing BIM models such as Autodesk Revit, Navisworks, and ArchiCAD for construction and infrastructure projects.
- Knowledge of Python for algorithm development and implementation.
- Proficient in HTML, CSS, and Bootstrap for front-end web development.

Interpersonal and Transferable Skills

- Excellent communication skills, both verbal and written, for effective collaboration and presentation of ideas.
- Strong teamwork and leadership abilities, demonstrated through successfully leading project teams.
- Problem-solving and critical thinking skills to identify innovative solutions and overcome challenges.

Language Skills

- Fluent in English (writing, reading, speaking, and listening).
- Proficient in Arabic (writing, reading, speaking, and listening).
- Fluent in Yoruba (writing, reading, speaking, and listening).

EDITORIAL WORK

Editorial board member of the Nature Scientific Reports Journal
International Program Committee Member for “1st International Conference on Agentic and Generative Techniques in Intelligent Computational Systems” - AGENTICS 2025
Editor for ANSHK 2024 Conference themed “Engaging Global Sustainability Issues through collaborative, interdisciplinary, and AI-driven research”

REVIEWER WORK

Reviewer for Automation in Construction (2025 – present)
Reviewer for Engineering Applications of Artificial Intelligence (2025 – present)
Reviewer for Journal of Asian Architecture and Building Engineering (2025 – present)
Reviewer for International Journal of Construction Management (2025 – present)
Reviewer for ASCE Journal of Construction Engineering and Management (2024 – present)
Reviewer for Case Studies in Construction Materials (2024 – present)
Reviewer for Cleaner Engineering and Technology (2024 – present)
Reviewer for Journal of Building Engineering (2024 – present)
Reviewer for International Journal of Computational Intelligence Systems (2024 – present)
Reviewer for Archives of Computational Methods in Engineering (2024 - present)
Reviewer for Environment, Development, and Sustainability (2024 - present)
Reviewer for Urban Water Journal (2024 - present)
Reviewer for Engineering Applications of Computational Fluid Mechanics (2024 – present)
Reviewer for Water Research Management (2023 – present)

PROFESSIONAL MEMBERSHIPS

Member, Institution of Civil Engineers: 93642167 (2022 – present)
Member, Institution of Engineering and Technology: 1100877380 (2020 – present)
Member, American Society of Civil Engineer: 11717875 (2019 – present)
Member, Environmental Water Resources Institute (2019 – present)

CERTIFICATIONS

Complete Machine Learning, NLP Bootcamp, MLOPS & Deployment (October 2024)
Udemy, USA
Complete Generative AI Course With Langchain and Huggingface (October 2024)
Udemy, USA
AWS Machine Learning Foundations (October 2022)
Udacity, USA
ALX-T Data Analyst Nanodegree (September 2022)
Udacity, USA
Introduction to SQL (June 2022)
DataCamp, USA
Python Programming Foundations (June 2022)
Udacity, USA
Lean Six Sigma White Belt Certification (February 2020)
Management and Strategy Institute, USA
Project Management Essentials Certification (August 2020)
Management and Strategy Institute, USA

SELECTED AI-RELATED PROJECTS

Compressive Strength of High-Performance Concrete Prediction App (2024)
- Developed an end-to-end framework for predicting compressive strength of high performance concrete by leveraging on ensemble learning techniques, feature optimization, and deployment.
- Deployed the model as a web application using Streamlit, accessible via this link.

Retrieval Augmented Generation (RAG) Powered LLM Model for Document Q &A (2024)
- Developed an end-to-end framework for answering questions based on users' uploaded documents using Retrieval-Augmented Generation (RAG) techniques and large language model (LLM).
- Utilized Groq and Google Generative AI embeddings to enhance document understanding and response accuracy.
- Deployed the model as a web-based application using Streamlit, accessible through this link here.

Causes of Water Pipe Failure Prediction Application (2024)
- Developed a web-based application that predicts the causes of water pipe failure by integrating XGBoost model with Bayesian Optimization algorithm.
- Implemented the model as a user-friendly web-based application using Streamlit, available here.

ADMINISTRATIVE & LEADERSHIP POSITIONS

Student Hall Coordinator, Homantin Hall, The Hong Kong Polytechnic University (09/2022 – 04/2024)

- Served as a member of the Student Hall Coordination team at Homantin Hall, The Hong Kong Polytechnic University.
- Acted as one of the Conciliators of Homantin Hall, facilitating open communication, resolving conflicts, and promoting a harmonious living environment for all residents.
- Contributed to the coordination and organization of hall activities, fostering a vibrant and inclusive community.
- Fulfilled the vital role of Treasurer in Homantin Hall, overseeing financial matters and ensuring the effective management of hall funds.

Osun State Ambassador for "Not Too Young to Lead" Initiative (06/2019 - Present)

- Representing Osun State as an ambassador for the "Not Too Young to Lead" Initiative, advocating for youth empowerment and effective leadership among young individuals.

Project Manager & General Secretary, Community Development Group of Road Safety Commission, Warri, Delta State (03/2019 – 12/2019)

- Successfully fulfilled the role of Project Manager and General Secretary in the Community Development Group of the Road Safety Commission in Warri, Delta State.
- Demonstrated effective project management and administrative skills.

Financial Secretary, Muslim Corpers Association of Nigeria, Delta State Chapter (03/2019 – 12/2019)

- Held the position of Financial Secretary for the Muslim Corpers Association of Nigeria, Delta State Chapter.
- Demonstrated financial management skills and contributed to the organization's activities and initiatives.

Co-founder, APM Foundation (11/2017 - Present)

- Co-founded the APM Foundation, an organization dedicated to providing crowdfunding assistance for health-related issues.
- Playing a pivotal role in assisting over 50 families by securing funds for medical treatments and health emergencies.
- Spearheading outreach efforts to raise awareness and gather support for the foundation's mission, ensuring continuous community engagement and support for those in need.

Head Prefect, Manbahu College of Science and Islamic Studies, Osogbo, Osun State (09/2009 – 08/2010)

- Displayed leadership qualities as the Head Prefect at Manbahu College of Science and Islamic Studies, Osogbo, Osun State, overseeing student affairs and maintaining discipline.

COMMUNITY/VOLUNTARY SERVICES

National Youth Service Corps, Nigeria (03/2019 – 12/2020)
General Secretary of Sozo Network Team, Warri, Delta State, Nigeria (03/2019 – 12/2020)
Member of Charity and Gender Equality Group Warri, Nigeria (03/2019 – 12/2020)
Co-founder of Assoutudeen Prophetic Medicine Foundation (11/2017 – present)
Member of Muslim Society Students of Nigeria (01/2014 – 12/2018)
Member of the United Nation Volunteers (01/2014 – 12/2018)

REFERENCES

Available upon request